Lab 4

In this lab, you will put all you have learned from cryptography lectures, particularly Diffie-Hellman and Hill Cipher code into action. You should now understand the concepts you need to secure communication between two parties (Bob and Alice) with an intruder Darth listening in and trying to listen in to the messages. The homework will focus on building an application that can send messages encrypted using hill cipher technique. As you may recall from Lab 3, the Custom encryption technique requires a numeric "N" which is used for encryption. In this homework, we will use the Diffie-Hellman key exchange to determine the value of the key which will be a shared secret between Bob and Alice and, use this key as the key for hill cipher technique. You will implement these techniques between two applications over an insecure UDP connection, which will help you think through how to securely send messages and ensure that they have not been eavesdropped on by intruders.

Learning Objectives:

- Construct an application that communicates over UDP
- Use the Diffie Hellman code that you will develop as part of this lab
- Use the Hill cipher Encryption technique that you will develop as part of this lab

<u>Task1: Diffie-Hellman algorithm</u>

The Diffie-Hellman algorithm was developed to create secure communications over a public network. For this lab, we will consider four variables that include P, G, A, B:

- **P:** One prime number; publicly available.
- **G:** A generator of P.
- **A:** A user (Alice) picks private values for A and B and use them to generate a key to exchange publicly with a second user (Bob).
- **B:** The second user (Bob) receives the key from Alice and uses it to generate a secret key; this gives both users the same secret key to encrypt.

To get a better understanding, review the following five steps:

1. Alice and Bob get public numbers P = 23, G = 9.
2. Each user selects a private key:
    - Alice selected a private key a = 4
    - Bob selected a private key b = 3
3. Each user computes public values:
    - Alice: x = (9^4 mod 23) = (6561 mod 23) = 6
    - Bob: y = (9^3 mod 23) = (729 mod 23) = 16
4. Alice and Bob exchange public numbers:
    - Alice receives public key y = 16
    - Bob receives public key x = 6
5. Alice and Bob compute symmetric keys:
    - Alice: ka = y^a mod p = 65536 mod 23 = 9
    - Bob: kb = x^b mod p = 216 mod 23 = 9

The completed process generates 9, which is the shared secret. Notice this value was never shared between the two parties.

Write a Python program that implements Diffie-Hellman method.

Hint: Python pow() Function The pow() function returns the value of x to the power of y. If a third parameter is present, it returns x to the power of y, modulus z.

Your DH program should have the three functions.

```
#Write Function 1 power(a,b,p) that returns a^b mod p using Python's
pow function. Handle special case if b == 1
def power(a, b, p):
    <Enter your code here>

#Write Function 2 that generates and returns a public key using P,G,
and a privateKey chosen by the sender
def dh_generatePublicKey(P,G,privateKey):
    <Enter your code here>

#Write Function 3 that generates and returns a private key using the
publicKey, privateKey and P chosen by the sender

def dh_generateSecretKey(publicKey, privateKey, P):
    <Enter your code here>
```

Your DH program should have the following main function

```
def main():
    P = 0; G = 0; x = 0; a = x;
    y = 0; b = 0;
    ka = 0; kb = 0;

    # Both the users will be agreed upon the public keys G and P
    P = 23; # A prime number P is taken
    print("The value of P:", P);

    G = 9; # A primitive root for P, G is taken
    print("The value of G:", G);

    # Alice will choose the private key a
    a = 4; # a is the chosen private key
    print("The private key a for Alice:", a);
    <Enter code here which calls the appropriate function from above
to generate public key for Alice>

    # Bob will choose the private key b
    b = 3; # b is the chosen private key
    print("The private key b for Bob:", b);
    <Enter code here which calls the appropriate function from above
to generate public key for Bob>

    # Generating the secret key after the exchange of keys
    <Enter code here which calls the appropriate function from above
to generate secret keys for both Alice and Bob. Test that these keys
match>
    print("Secret key for the Alice is:", ka);
    print("Secret Key for the Bob is:", kb);

if __name__ == '__main__':
```

```
main()
```

Note, while the above main program has specific values of P,G and private keys for Bob and Alice, your program should work for any user-specified values.

Task 2: Hill Cipher

In the field of cryptography, the substitution cipher simply substitutes one letter in the alphabet for another based upon a crypto variable. The substitution involves shifting positions in the alphabet. This includes the Caesar Cipher and ROT-13. Examine the following example:

- Plaintext: WE HOLD THESE TRUTHS TO BE SELF-EVIDENT, THAT ALL MEN ARE CREATED EQUAL.
- Ciphertext: ZH KROG WKHVH WUXWKV WR EH VHOI-HYLGHQW, WKDW DOO PHQ DUH FUHDWHG HTXDO.

In lab3, you implemented a custom algorithm for Caesar Cipher. Some other types of substitution ciphers include the Vigenere Cipher, and the Playfair Cipher. The Hill 2x2 cipher is a polygraphic substitution cipher based on linear algebra. The inventor of this cipher, Lester S. Hill, created the cipher in 1929. The cipher uses matrices and matrix multiplication to mix the plaintext to produce the ciphertext. This cipher is based on a branch of mathematics known as number theory. The key used here is HILL, which corresponds to the numbers 7, 8, 11, and 11. For this task, you will develop two Python programs.

1. A function to perform encryption of a plaintext input to cipher text using the Hill 2x2 technique.
2. A function to perform decryption of a plaintext input to cipher text using the Hill 2x2 technique.

A skeleton code for the encryption/decryption algorithm is shown below: (The below code is just for your reference, you can write your own). Do not hard code. Use “HILL” as the key for Hill Cipher technique.

```
import sys
import numpy as np

def cipher_encryption(plain, key):

    # Handle condition if the message length is odd.
    #<Enter code here>

    # Convert msg to matrices
    #Enter code here>

    # Convert key to 2x2
    #Enter code here>

    print (key2d)

    # checking validity of the key
    # finding determinant
    #Enter code here>
```

```
    # finding multiplicative inverse and implementing steps to encrypt
text
    #<Enter code here>

    print("Encrypted text: {}".format(encryp_text))
    return encryp_text


def cipher_decryption(cipher, key):

    # Handle condition if the message length is odd.
    #<Enter code here>

    # Convert msg to matrices
    #<Enter code here>

    # Convert key to 2x2
    #<Enter code here>

    # finding determinant
    #<Enter code here>

    # finding multiplicative inverse
    #<Enter code here>

    # find transpose
    # find minor
    #<Enter code here>

    #changing signs
    #<Enter code here>

    # multiplying multiplicative inverse with adjugate matrix
    #<Enter code here>

    # Calculate modulo
    #<Enter code here>

    # Convert cipher to plaintext
    #<Enter code here>

    print("Decrypted text: {}".format(decryp_text))


Example:
plaintext = "Secret Message"
plaintext = plaintext.upper().replace(" ","")
key = "hill"
key = key.upper().replace(" ","")
ciphertext = cipher_encryption(plaintext, key)
cipher_decryption(ciphertext, key)
```

Task 3: Constructing a Plaintext Communications Application

The architecture for the application includes a server file (alice.py), a client file (bob.py), an intruder (darth.py) and a helper file for shared components used by all the files share many of the same components. Alice first shares the shared public key with Bob. This application assumes that Bob has already shared his public key with Alice. We are assuming this to keep the application simple for the homework. In the real world, Bob will share his key with Alice over the channel. Alice will then compute the shared Secret using Bob's public key and encrypt the message before sending it over the channel. Bob will also compute the shared Secret using Alice's key and decrypt the message. To iterate, to keep this example simple, we will only do one-way communication. The same techniques used in this chapter can be used between any applications that transmit data over UDP.

Creating Code for Alice.py

Our first step is to create an application (alice.py) that will use UDP sockets to send messages to Bob. She is not expecting to send messages to Darth, but Darth is eavesdropping on the messages. A skeleton code for alice.py is provided to you with instructions on where to introduce your code.

Creating Code for Bob.py and Darth.py

Bob.py and Darth.py will largely stay the same with the exceptions of the private key for each of them. A skeleton code for bob.py and darth.py is provided to you with instructions on where to introduce your code.

Creating the Helper File

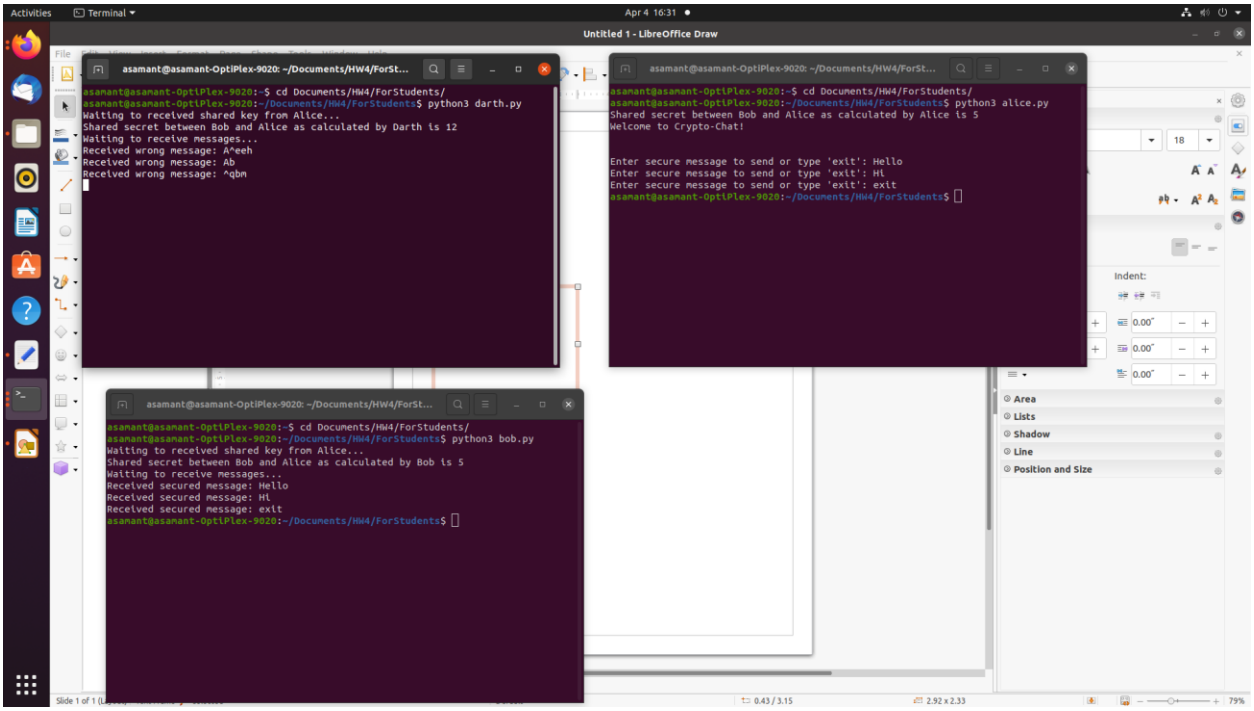
To keep the code as consistent as possible between each step in our application, we will create some system components that are used in alice.py, bob.py and darth.py. These components are saved in the file lab4_shared.py. The below figure shows a screenshot of the three Terminal Windows (Command Windows) running the three programs bob.py, alice.py, and darth.py. Please note that the listeners bob.py and darth.py should be run first before running alice.py. Also, while alice.py and bob.py programs will properly exit when they receive the text "exit", darth.py will not exit (as per current logic). You will have to manually terminate the window.

Note: Students running the code on Linux machines will be able to run all three programs together. For Windows OS, the following command

```
UDPSock.setsockopt(SOL_SOCKET, SO_REUSEPORT, 1)
```

does not work on Win7 and higher. Students can either find a different way for both bob.py and darth.py to listen to the same port or they can first run alice.py and bob.py together and then run alice.py and darth.py together. A quick demo is shown here:

https://www.youtube.com/watch?v=j-ggSi0Wj7E



What to submit, a zip file containing below files:

1. Updated version of alice.py
2. Updated version of bob.py
3. Updated version of darth.py
4. Updated version of lab4_shared.py
5. dh.py
6. screenshot of command windows showing the communication between Alice, Bob and Darth

Assumptions:

1. The input text (message being sent) does not include any spaces between the words.

Rubric

Submission Artifact	Points
Updated version of alice.py	20
Updated version of bob.py	20
Updated version of darth.py	20
Updated version of lab4_support.py	20
Screenshot of command windows	20